Ericsson PE Device Troubleshooting SOP for QoS / Congestion Event on WFD-PE-02 at Whitefield POP

Node: WFD-PE-02 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_8075

Description

The WFD-PE-02 device, located at the Whitefield POP, running Ericsson's CLI dialect, is experiencing a critical QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) as indicated by Alarm Code ALARM_8075. This failure domain occurs due to the PE's role in managing Quality of Service (QoS) and handling congestion events, which are inherent in the protocol mechanics of Multiprotocol Label Switching (MPLS) and the hardware design of the device. In specific, the PE device's control plane, responsible for QoS policy management, encounters microbursts caused by excessive packet loss or interface congestion, leading to queue starvation and subsequent tail drops. The Ericsson device's hardware, consisting of the Route Processor (RP) and Line Cards (LCs), also plays a crucial role in this scenario. The RP processes QoS policies and makes forwarding decisions, while the LCs handle data plane operations, including packet forwarding and queuing. As a result, when the RP encounters congestion or microbursts, it may not be able to handle QoS policy management efficiently, leading to the observed behavior.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and that remote hands are dispatched to the site.
- Check the redundancy status of the links connected to the WFD-PE-02 device.
- Gather relevant logs and performance data from the device to aid in troubleshooting.

Potential Root Causes

Cause	Probability
High CPU Utilization on the Route Processor (RP)	High
Excessive Packet Loss or Interface Congestion	Medium
Misconfigured or Inadequate QoS Policies	Low

Diagnostic Commands & Expected Outputs

```
Command: show cpu cpu-all, Expected Output: CPU Utilization %, Description: This command displays the CPU utilization of the Route Processor (RP), which can help identify if the high CPU utilization is causing the issue.

Command: show interface counters, Expected Output: Interface Counter Statistics, Description: This command displays the interface counter statistics, which can help identify excessive packet loss or interface congestion.
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```
Command: show qos policy, Expected Output: QoS Policy Configuration, Description: This command displays the QoS policy configuration, which can help identify misconfigured or inadequate QoS policies.
```

Resolution Steps

Step 1: Step: Step 1: Reduce CPU Utilization on the Route Processor (RP), Command: process cpu-queue max-threads, Description: This command reduces the CPU utilization on the Route Processor (RP) by limiting the number of threads.

Step 2: Step: Step 2: Adjust QoS Policies, Command: qos policy modify, Description: This command adjusts the QoS policies to ensure they are adequate for the network traffic.

Step 3: Step: Step 3: Check and Clear Interface Counters, Command: clear interface counters, Description: This command clears the interface counter statistics to prevent false positives.

Post-Check Verification

```
Command: show cpu cpu-all, Expected Output: CPU Utilization %, Description: Verify that CPU utilization has been reduced.

Command: show interface counters, Expected Output: Interface Counter Statistics, Description: Verify that interface counter statistics have improved.

Command: show qos policy, Expected Output: QoS Policy Configuration, Description: Verify that QoS policies have been adjusted correctly.
```

Roll-back Procedures

- Step: Step 1: Restore Original QoS Policy Configuration, Command: qos policy restore, Description: Restore the original QoS policy configuration in case the adjustments made did not resolve the issue.

- Step: Step 2: Verify CPU Utilization and Interface Counters, Command: show cpu cpu-all, Description: Verify that CPU utilization and interface counters have returned to normal.